

Department of Physics, FNSPE, CTU in Prague

02YMECH - Mechanics

Midterm 1

7.11.2025 / 14:00 - 16:00

Q1: Three point particles collide instantaneously at the origin with no external force. Their masses and pre-collision velocities are

$$m_1 = 1 \text{ kg}, \quad m_2 = 2 \text{ kg}, \quad m_3 = 3 \text{ kg},$$

$$\vec{v}_{1i} = (2, 0, 1) \text{ m/s}, \quad \vec{v}_{2i} = (-1, 1, 0) \text{ m/s}, \quad \vec{v}_{3i} = (0, -2, 3) \text{ m/s}.$$

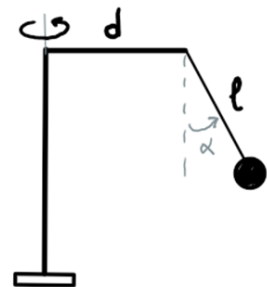
Immediately *after* the collision, particle 1 moves along $\hat{i}$, particle 2 along $-\hat{j}$, and particle 3 along $+\hat{k}$.

(a) Determine the velocities after the collision: $\vec{v}_{1f}, \vec{v}_{2f}, \vec{v}_{3f}$. — *0.6 points*

(b) Determine whether the collision is elastic or not. — *0.4 points*

Q2: A projectile with a mass m strikes a wall with a velocity v_1 and exits from the other side with a velocity v_2 after penetrating the wall of thickness d . What was the average force F acting on the projectile inside the wall? — *1 point*

Q3: A mathematical pendulum with a length of l is attached to the arm of a rotating carousel at a distance d from its axis. The pendulum is deflected from its vertical position by an angle α . Calculate the period of rotation for the carousel. — *1 point*



Q4: A particle of mass m can move without friction along the x -axis. At time $t = 0$, it is at rest at position $x = 0$. A force $F_x(t) = k \cos(\omega t)$ acts on it, where k and ω are constants.

(a) Determine the acceleration $a(t)$, the velocity $v(t)$, the displacement $x(t)$. — *0.6 points*

(b) Determine the power delivered by the force at an arbitrary time t . — *0.4 points*